

Integrating Real-Time ECG Data into Virtual Reality for Enhanced Medical Training

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Figure 1: A VR hospital operating room featuring medical equipment and a healthcare professional using a monitoring device, designed for training healthcare personnel in a realistic environment to practice and refine medical procedures.

ABSTRACT

Healthcare trainees often encounter challenges such as high-pressure scenarios, risks to patient safety, limited exposure to specific cases, and restricted access to advanced tools. Virtual Reality (VR) addresses these issues by offering a controlled, immersive environment for repeated, risk-free practice. This work presents a VR training system that allows trainees to engage with real-time ECG data, observe live procedures remotely, and provide feedback. This approach ensures patient anonymity with encrypted data, the system connects theoretical learning with practical application, demonstrating its effectiveness in improving clinical skills and providing a flexible, secure training option for settings with limited patient access.

Index Terms: Computing methodologies—Computer graphics—Graphics systems and interfaces—Virtual reality; Security and privacy Human—Societal aspects of security—Privacy Usability in security and privacy—

1 INTRODUCTION

In medical training, particularly cardiology, practical experience is vital to develop ECG analysis skills, a key tool for monitoring cardiac conditions [2]. Traditional methods like lectures and static simulations lack the interactivity needed to build confidence, a gap emphasized during the COVID-19 pandemic when social distancing restricted hands-on training, highlighting the need for innovative solutions [1]. Therefore, situations where traditional training methods are insufficient, emphasize the need for a system that ensures secure, real-time data streaming in a safe environment for skill development.

The introduction of VR technology has brought about a paradigm shift in the field of medical education and training, offering safe, immersive environments for repeated practice without patient risk

[5] [3]. Although VR is widely used in areas such as surgery and emergency simulations, its application in ECG training remains underexplored. Integrating real-time physiological data into VR could significantly enhance realism and clinical relevance. In this regard, it is interesting to ask what challenges and advantages the safe integration of ECG data in virtual reality environments poses for medical training.

This paper introduces a VR-based training platform that integrates an ECG sensor with Unity for real-time data streaming and visualization. It allows trainees to practice ECG analysis in real-time while providing feedback to the supervising physician, fostering hands-on skills in a controlled environment. Additionally, it addresses data privacy through encrypted ECG transmission, ensuring security against cyber threats. This solution bridges the gap between theory and practice, offering a flexible and secure training method adaptable to conditions like pandemics. The approach enhances medical training through real-time interactive learning, secure ECG data integration, and remote collaboration, providing a more immersive, versatile, and accessible experience than traditional methods.

2 SECURE REAL-TIME ECG DATA ACQUISITION, PROCESSING, AND TRANSMISSION

The ProtoCentral MAX30001 Breakout Board ¹ is the sensor used for ECG data acquisition, featuring the MAX30001 AFE chip for single-lead monitoring, 18-bit ADC resolution, ± 650 mV motion artifact tolerance, and built-in R-R detection, enabling low-noise, real-time cardiac analysis with ESP32 control.

The proposed system uses a hybrid TCP-UDP approach to securely transmit ECG sensor data to a VR headset. Although hybrid TCP/UDP protocols are not novel, they address the unique needs of VR medical training by combining TCP's reliability for establishing the connection with UDP's low latency for transmitting encrypted data in real time. The ECG data are transmitted directly to the Meta Quest 3 standalone headset, bypassing Unity to avoid additional pro-

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¹https://github.com/Protocentral/protocentral_max30001

cessing delays. This ensures seamless integration of real-time data in the VR environment. This strategy balances security, reliability, and efficiency, meeting the demands of encrypted medical data in interactive VR environments [4]. Figure 2 illustrates a data flow linking trainees, sensor installation, encrypted data transmission, and visualization in a 3D virtual environment. The main steps are:

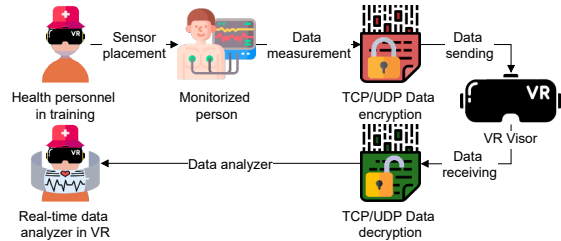


Figure 2: ECG Data processing for personnel training in VR.

Health personnel training: Trainees use VR controllers to interact with a 3D environment, placing sensors, interpreting real-time ECG signals, and providing feedback, simulating common medical tasks for practical skill transfer.

Monitored person: Sensors placed on a real patient collect heart rate data in real-time, processed by a connected system.

TCP/UDP Data encryption: ECG data is encrypted for privacy and sent securely over a TCP/UDP network.

VR Visor: Trainees' VR visor receives the data and assembles the 3D scenario for subsequent visualization.

TCP/UDP Data decryption: The server decrypts the received data using secure keys to reconstruct it.

Real-time data analyzer in VR: The data is transformed into 3D graphs, allowing trainees to analyze heart behavior.

3 VIRTUAL REALITY TRAINING SCENE

The VR training environment immerses medical trainees in a realistic scenario where they actively engage with real-time ECG data (Figure 1) while a supervising physician manages the physical patient. Using real-time data replicates high-pressure scenarios, fostering critical thinking and skill development in real-world conditions. The system supports real-time monitoring for detecting critical cardiac events, such as arrest, infarction, unstable angina, or tachycardias, which trainees analyze to provide feedback and notify their opinion in real time to the supervising physician. The system also measures trainees' stress levels using VR controller feedback, rating them on a scale from 0 (no stress) to 5 (critical stress). This data supports both patient evaluation and assessment of trainees' analytical skills. All interactions and feedback are logged, contributing to trainees' course evaluations and enhancing the training experience through active collaboration, deeper learning, and robust clinical and educational assessments.

4 RESULTS

Real-time ECG data was successfully transmitted to a VR training environment, enabling smooth and accurate visualization at 30 FPS with a sampling frequency of 200 Hz. This system allows trainees to analyze live patient data, interact with realistic scenarios, and receive immediate feedback, providing an immersive and scalable training experience. As shown in Figure 3, the top figure illustrates the ECG waveform captured in a TFT display during a live consultation, while the bottom figure demonstrates its representation on a virtual monitor within the VR environment.

5 CONCLUSIONS

This work demonstrates that integrating real-time encrypted ECG data into a VR environment provides a secure and scalable solution



Figure 3: **Top:** Real-time ECG data viewed by the physician during a consultation. **Bottom:** The same data displayed in the VR environment on a virtual monitor, enabling trainees to analyze the patient's heart activity in an immersive scenario.

for medical training. By replicating high-pressure scenarios, the system fosters critical thinking and practical skills while ensuring low-latency and secure data transmission through a hybrid TCP/UDP protocol. Future work will focus on integrating digital twins for predictive analysis, enabling autonomous detection capabilities, and conducting user studies to evaluate its impact on medical education.

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